

Laplace Transform Theory

1 Introduction

Let

$$\Psi_b(t) = \Psi_{b0} \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}. \quad (1)$$

It follows that

$$\bar{\Psi}_b(g) = \Psi_{b0} \int_0^\infty e^{-gt} dt = \frac{\Psi_{b0}}{g}. \quad (2)$$

Now, the Laplace transformed reconnected magnetic flux is

$$\bar{\Psi}_0(g) = \frac{E_{sb} F_s(\hat{g}) \bar{\Psi}_b(g)}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}} = E_{sb} \Psi_{b0} \frac{1}{g} \frac{F_s(\hat{g})}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}, \quad (3)$$

where $\hat{g} = S^{1/3} g \tau_H$. The reconnected flux is

$$\Psi_0(t) = \frac{1}{2\pi i} \int_c \bar{\Psi}_0(g) e^{gt} dg, \quad (4)$$

where c is the Bromwich contour. Thus, we obtain

$$\Psi_0(\hat{t}) = \frac{1}{2\pi i} \int_c \bar{\Psi}_0(\hat{g}) e^{\hat{g}\hat{t}} d\hat{g}, \quad (5)$$

where $\hat{t} = t/(S^{1/3} \tau_H)$ and

$$\bar{\Psi}_0(\hat{g}) = E_{sb} \Psi_{b0} \frac{1}{\hat{g}} \frac{F_s(\hat{g})}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}. \quad (6)$$

2 Non-Constant- ψ Regimes

We shall assume that $|S^{1/3} \hat{\Delta}_s| \gg |E_{ss}|$ in the non-constant- ψ regimes, so

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{E_{sb} \Psi_{b0}}{S^{1/3}} \frac{1}{\hat{g}} \frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})}. \quad (7)$$

2.1 Inertial Regime

In the inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}, \quad (8)$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)]}, \quad (9)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi} \frac{2}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}} \quad (10)$$

Thus,

$$\begin{aligned} \bar{\Psi}_0(\hat{g}) &= \frac{2}{\pi} \frac{E_{sb} \bar{\Psi}_{b0}}{S^{1/3} i (Q_E + Q_e)} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i (Q_E + Q_e)} \right] \\ &\quad \times \frac{1}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}. \end{aligned} \quad (11)$$

Now, if

$$\bar{\Psi}_0(\hat{g}) = \frac{1}{\hat{g} + a} \bar{H}(\hat{g}) \quad (12)$$

then (ERD:¹ 4.1.22)

$$\Psi_0(\hat{t}) = \int_0^{\hat{t}} e^{a(\hat{t}' - \hat{t})} H(\hat{t}') d\hat{t}'. \quad (13)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}} \quad (14)$$

¹Erdélyi, et al. *Tables of Integral Transforms*, vol. I.

$$= \frac{1}{(\hat{g}^2 + \alpha \hat{g} + \beta)^{1/2}}, \quad (15)$$

where

$$\alpha = i(2Q_E + Q_i), \quad (16)$$

$$\beta = -Q_E(Q_E + Q_i). \quad (17)$$

Note that

$$\beta - \frac{1}{4}\alpha^2 = -Q_E(Q_E + Q_i) + \frac{1}{4}(2Q_E + Q_i)^2 = \frac{Q_i^2}{4}. \quad (18)$$

However (ERD: 5.3.34),

$$H(\hat{t}) = e^{-\alpha \hat{t}/2} J_0[(\beta - \alpha^2/4)^{1/2} \hat{t}] \quad (19)$$

$$= e^{-i(Q_E + Q_i/2)\hat{t}} J_0\left(\frac{|Q_i|}{2} \hat{t}\right). \quad (20)$$

Thus,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{2}{\pi} \frac{E_{sb} \Psi_{b0}}{S^{1/3} i(Q_E + Q_e)} \int_0^{\hat{t}} \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})}\right] \\ &\times e^{-i(Q_E + Q_i/2)\hat{t}'} J_0\left(\frac{|Q_i|}{2} \hat{t}'\right) d\hat{t}'. \end{aligned} \quad (21)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{2}{\pi} \frac{E_{sb} \Psi_{b0}}{S^{1/3}} \frac{\hat{t}^2}{2}. \quad (22)$$

At large $\hat{t}$,

$$\begin{aligned} \Psi_0(\hat{t}) &\simeq \frac{2}{\pi} \frac{E_{sb} \Psi_{b0}}{S^{1/3} i(Q_E + Q_e)} \int_0^{\infty} \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})}\right] \\ &\times e^{-i(Q_E + Q_i/2)\hat{t}'} J_0\left(\frac{|Q_i|}{2} \hat{t}'\right) d\hat{t}'. \end{aligned} \quad (23)$$

But (GR:² 6.621.4)

$$\int_0^{\infty} e^{-\alpha x} J_0(\beta x) dx = \frac{1}{\sqrt{\alpha^2 + \beta^2}}, \quad (24)$$

²I.S. Gradshteyn and I.M. Ryzhik, *Table of Integrals, Series, and Products*, Corrected and Enlarged Edition

so

$$\Psi_{b0}(\hat{t}) \simeq -\frac{2}{\pi} \frac{E_{sb} \Psi_{b0}}{S^{1/3} (Q_E + Q_i)} \left[\frac{1}{\sqrt{Q_E (Q_E + Q_i)}} - \frac{e^{-i(Q_E + Q_e) \hat{t}}}{\sqrt{Q_e (Q_e - Q_i)}} \right]. \quad (25)$$

2.2 Viscous-Inertial Regime

In the Viscous-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{1/4} P_\varphi^{1/4}}, \quad (26)$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2} P_\varphi^{1/2}}, \quad (27)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{1}{\pi^{1/2}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4} P_\varphi^{1/4}}. \quad (28)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{1}{\pi^{1/2}} \frac{E_{sb} \bar{\Psi}_{b0}}{S^{1/3} P_\varphi^{1/4}} \frac{1}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}. \quad (29)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}, \quad (30)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(5/4)} \hat{t}^{1/4} e^{-i(Q_E + Q_e) \hat{t}}. \quad (31)$$

Hence,

$$\Psi_0(\hat{t}) = \frac{1}{\pi^{1/2} \Gamma(5/4)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} P_\varphi^{1/4}} \int_0^{\hat{t}} \hat{t}'^{1/4} e^{-i(Q_E + Q_e) \hat{t}'} d\hat{t}' \quad (32)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{1}{\pi^{1/2} \Gamma(5/4)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} P_\varphi^{1/4}} \frac{\hat{t}^{5/4}}{5/4}. \quad (33)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{1}{\pi^{1/2} \Gamma(5/4)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} P_\varphi^{1/4}} \int_0^\infty \hat{t}'^{1/4} e^{-i(Q_E + Q_e) \hat{t}'} d\hat{t}' \quad (34)$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} e^{-i q x} dx = \frac{\Gamma(\nu)}{(i q)^\nu}, \quad (35)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{1}{\pi^{1/2}} \frac{E_{sb} \Psi_{b0}}{S^{1/3} P_\varphi^{1/4}} \frac{1}{[i(Q_E + Q_e)]^{5/4}}. \quad (36)$$

2.3 Diffusive-Inertial Regime

In the Diffusive-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi D}{[\hat{g} + i(Q_E + Q_e)]^{1/2} \iota_e^{1/2} P_\perp^{1/2}}, \quad (37)$$

$$F_s(\hat{g}) = -\frac{2 D^2}{[\hat{g} + i(Q_E + Q_e)]^2 \iota_e P_\perp}, \quad (38)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2} \iota_e^{1/2} P_\perp^{1/2} D^{-1}}. \quad (39)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{2}{\pi} \frac{E_{sb} \bar{\Psi}_{b0}}{S^{1/3} \iota_e^{1/2} P_\perp^{1/2} D^{-1}} \frac{1}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2}}. \quad (40)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2}}, \quad (41)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(3/2)} \hat{t}^{1/2} e^{-i(Q_E + Q_e) \hat{t}}. \quad (42)$$

Hence,

$$\Psi_0(\hat{t}) = \frac{2}{\pi \Gamma(3/2)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} \iota_e^{1/2} P_\perp^{1/2} D^{-1}} \int_0^{\hat{t}} \hat{t}'^{1/2} e^{-i(Q_E + Q_e) \hat{t}'} d\hat{t}' \quad (43)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{2}{\pi \Gamma(3/2)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} \iota_e^{1/2} P_\perp^{1/2} D^{-1}} \frac{\hat{t}^{3/2}}{3/2}. \quad (44)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) = \frac{2}{\pi \Gamma(3/2)} \frac{E_{sb} \Psi_{b0}}{S^{1/3} \iota_e^{1/2} P_{\perp}^{1/2} D^{-1}} \int_0^{\infty} \hat{t}'^{1/2} e^{-i(Q_E + Q_e) \hat{t}'} d\hat{t}' \quad (45)$$

But (GR: 3.381.5),

$$\int_0^{\infty} x^{\nu-1} e^{-i q x} dx = \frac{\Gamma(\nu)}{(i q)^{\nu}}, \quad (46)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{2}{\pi} \frac{E_{sb} \Psi_{b0}}{S^{1/3} \iota_e^{1/2} P_{\perp}^{1/2} D^{-1}} \frac{1}{[i(Q_E + Q_e)]^{3/2}}. \quad (47)$$

3 Constant- ψ Regimes

We shall not assume that $|S^{1/3} \hat{\Delta}_s| \gg |E_{ss}|$ in the constant- ψ regimes, which are characterized by $F_s(\hat{g}) = 1$, so

$$\bar{\Psi}_0(\hat{g}) \simeq E_{sb} \Psi_{b0} \frac{1}{\hat{g}} \frac{1}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}. \quad (48)$$

3.1 Viscous-Resistive Regime

In the Viscous-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3} \pi \Gamma(5/6)}{\Gamma(1/6)} [\hat{g} + i(Q_E + Q_e)] P_{\varphi}^{1/6}, \quad (49)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{1}{\hat{g}} \frac{\epsilon}{[\hat{g} + i(Q_E + Q_e)] + \epsilon}, \quad (50)$$

where

$$\epsilon = \frac{\Gamma(1/6)}{6^{3/2} \pi \Gamma(5/6)} \frac{(-E_{ss})}{S^{1/3} P_{\varphi}^{1/6}}. \quad (51)$$

Note that $\epsilon > 0$. Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i(Q_E + Q_e) + \epsilon} \right], \quad (52)$$

giving

$$\Psi_0(\hat{t}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon} \left[1 - e^{-i(Q_E + Q_e)\hat{t}} e^{-\epsilon\hat{t}} \right]. \quad (53)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \epsilon \hat{t}. \quad (54)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon}. \quad (55)$$

3.2 Diffusive-Resistive Regime

In the Diffusive-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [\hat{g} + i(Q_E + Q_e)] \iota_e^{1/4} P_\perp^{1/4} D^{-1/2}, \quad (56)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{1}{\hat{g}} \frac{\epsilon}{[\hat{g} + i(Q_E + Q_e)] + \epsilon}, \quad (57)$$

where

$$\epsilon = \frac{\Gamma(1/4)}{2\pi \Gamma(3/4)} \frac{(-E_{ss})}{S^{1/3} \iota_e^{1/4} P_\perp^{1/4} D^{-1/2}}. \quad (58)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i(Q_E + Q_e) + \epsilon} \right], \quad (59)$$

giving

$$\Psi_0(\hat{t}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon} \left[1 - e^{-i(Q_E + Q_e)\hat{t}} e^{-\epsilon\hat{t}} \right]. \quad (60)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \epsilon \hat{t}. \quad (61)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{i(Q_E + Q_e) + \epsilon}. \quad (62)$$

3.3 Semi-Collisional Regime

In the Semi-Collisional regime,

$$\hat{\Delta}_s(\hat{g}) = \pi [\hat{g} + \mathrm{i}(Q_E + Q_e)] (\hat{g} + \mathrm{i}Q_E)^{1/2} D^{-1}, \quad (63)$$

so

$$\hat{\Psi}_{b0}(\hat{g}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{1}{\hat{g}} \frac{\epsilon}{[\hat{g} + \mathrm{i}(Q_E + Q_e)] (\hat{g} + \mathrm{i}Q_E)^{1/2} + \epsilon}, \quad (64)$$

where

$$\epsilon = \frac{1}{\pi} \frac{(-E_{ss})}{S^{1/3} D^{-1}}. \quad (65)$$

If we neglect ϵ in the denominator then

$$\hat{\Psi}_{b0}(\hat{g}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{1}{\hat{g}} \frac{\epsilon}{[\hat{g} + \mathrm{i}(Q_E + Q_e)] (\hat{g} + \mathrm{i}Q_E)^{1/2}}, \quad (66)$$

giving

$$\hat{\Psi}_{b0}(\hat{g}) \simeq \frac{E_{sb} \Psi_{b0}}{(-E_{ss}) \mathrm{i}(Q_E + Q_e)} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + \mathrm{i}(Q_E + Q_e)} \right] \frac{\epsilon}{(\hat{g} + \mathrm{i}Q_E)^{1/2}}. \quad (67)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + \mathrm{i}Q_E)^{1/2}}, \quad (68)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(1/2)} \hat{t}^{-1/2} \mathrm{e}^{-\mathrm{i}Q_E \hat{t}}. \quad (69)$$

Hence,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{E_{sb} \Psi_{b0} \epsilon}{(-E_{ss}) \Gamma(1/2) \mathrm{i}(Q_E + Q_e)} \\ &\times \int_0^{\hat{t}} \left[1 - \mathrm{e}^{\mathrm{i}(Q_E + Q_e)(\hat{t}' - \hat{t})} \right] \hat{t}'^{-1/2} \mathrm{e}^{-\mathrm{i}Q_E \hat{t}'} d\hat{t}'. \end{aligned} \quad (70)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss}) \Gamma(1/2)} \frac{\epsilon \hat{t}^{3/2}}{3/4}. \quad (71)$$

At large $\hat{t}$,

$$\begin{aligned}\Psi_0(\hat{t}) &= \frac{E_{sb} \Psi_{b0} \epsilon}{(-E_{ss}) \Gamma(1/2) \mathrm{i} (Q_E + Q_e)} \\ &\times \int_0^\infty \left[1 - \mathrm{e}^{\mathrm{i} (Q_E + Q_e) (\hat{t}' - \hat{t})} \right] \hat{t}'^{-1/2} \mathrm{e}^{-\mathrm{i} Q_E \hat{t}'} d\hat{t}'.\end{aligned}\quad (72)$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} \mathrm{e}^{-\mathrm{i} q x} dx = \frac{\Gamma(\nu)}{(\mathrm{i} q)^\nu}, \quad (73)$$

It follows that

$$\Psi_0(\hat{t}) = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})} \frac{\epsilon}{\mathrm{i} (Q_E + Q_e)} \left[\frac{1}{(\mathrm{i} Q_E)^{1/2}} - \frac{\mathrm{e}^{-\mathrm{i} (Q_E + Q_e) \hat{t}}}{(-\mathrm{i} Q_e)^{1/2}} \right]. \quad (74)$$